Do Now · retrieval

Bubble sort: trace and implement

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

For CountDigits(582), list the calls and give the returned digit count.

Q2

How many times does a loop with i = 0; i < 4; i++ run?

Q3

For A=true and B=false, what is A && B?

Q4

After swapping neighbouring values 7 and 3 in [7,3,5], what is the array?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Compare neighbours — Compare positions j and j+1. Swap only if the earlier value is greater.

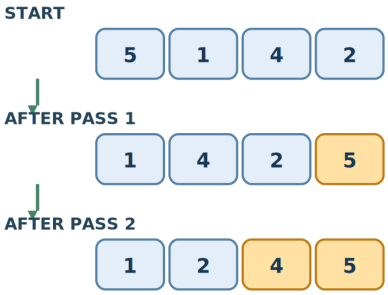
Core idea 2

Finish one pass — The greatest unsorted value moves to the right-hand end of the unsorted section.

Core idea 3

Repeat less work — Each completed pass fixes one final position; an entire pass with no swaps proves the array is sorted.

Key visual · complete concept



New Learning · applying the ideas

Vocabulary · use the correct technical terms

pass — One traversal of the remaining unsorted section.

adjacent — At neighbouring indices.

swap — Exchange two stored values.

comparison — Test used to determine whether a swap is required.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

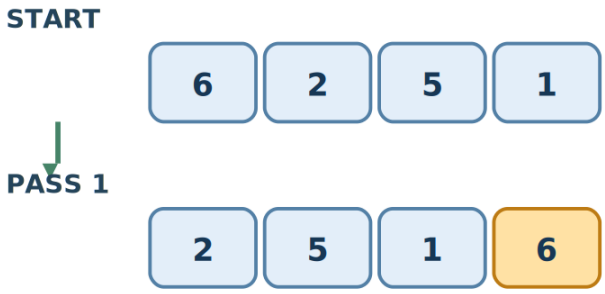
Problem

Trace a complete pass of ascending bubble sort on [6,2,5,1].

Follow the worked method

1. Predict first — Compare (6,2), (6,5), then (6,1).
2. Model solution — After swaps: [2,6,5,1] → [2,5,6,1] → [2,5,1,6]. The last value is fixed.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first
Which adjacent pair is compared first?

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Show one left-to-right pass on [7,3,6,2].

Question 2

Show two complete passes on [7,3,6,2].

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Why can the early-exit optimisation finish immediately on [1,2,3,4]?

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Bubble sort always needs to make at least one swap.

Use key vocabulary from memory

1. Explain pass in your own words.

2. Explain adjacent in your own words.

3. Explain swap in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.